

Research Progress Report (Weeks 1–2)

Reporting window: January 21–February 3, 2026 (status as of February 3, 2026)

Scope. This report documents the process, artifacts, and deliverables completed during Week 1 (data acquisition, preprocessing, baseline evaluation, and an LSTM residual-correction baseline) and Week 2 (advanced models, ablations, and tuning). All dates refer to the original plan window (January 21–February 3, 2026), and all items below reflect the current state of the codebase as of February 3, 2026.

1 Week 1 (January 21–January 27, 2026): Data, Baselines, and LSTM

Goal. Establish an end-to-end pipeline from raw data to reproducible baseline metrics, and complete initial LSTM residual-correction runs.

1.1 Site selection and mapping lock

Process.

- Finalized four study sites and locked required metadata fields in a dedicated configuration.
- Constrained `config/master_study_sites.py` to the four gauges used in the study.

Deliverable. `config/sites.yaml` with fields `usgs_id`, `comid`, `name`, `lat`, `lon`, `watershed`, `regulated_flag`, and `upstream_of`.

1.2 Retrospective data acquisition

Process.

- National Water Model (NWM) v2.x retrospective hourly streamflow retrieved per COMID.
- USGS archival hourly discharge retrieved per gauge.
- ERA5 hourly meteorology retrieved per site, restricted to precipitation, temperature, and soil moisture.

Artifacts.

- NWM v2 retrospective CSVs under `data/raw/nwm_v2/...`
- USGS CSVs under `data/raw/usgs/`
- ERA5 CSVs under `data/raw/era5/`

Scripts.

- `data_acquisition_scripts/nwm.py`
- `data_acquisition_scripts/usgs.py`
- `data_acquisition_scripts/era5.py` (restricted to the desired variables and output columns)

1.3 Per-site preprocessed datasets

Process.

- Constructed a combined training parquet by joining NWM, USGS, and ERA5 sources.
- Generated per-site processed tables with fields `timestamp`, `Q_nwm`, `Q_obs`, `error`, meteorological covariates, and `hour_of_day`.
- Implemented missing-value handling for meteorological variables in the dataset builder.

Scripts.

- `modeling/build_training_dataset.py`
- `scripts/build_processed_parquets.py`
- `scripts/split_training_by_site.py`

Deliverables.

- Combined parquet: `data/processed/watauga_cluster_2010_2020_all_sites.parquet`
- Per-site parquets:
 - `data/processed/watauga_cluster_2010_2020_site_03161000.parquet`
 - `data/processed/watauga_cluster_2010_2020_site_03164000.parquet`
 - `data/processed/watauga_cluster_2010_2020_site_03479000.parquet`
 - `data/processed/watauga_cluster_2010_2020_site_03486000.parquet`

1.4 Baseline evaluation (raw NWM vs. USGS)

Process. Computed baseline performance per site using correlation coefficient (CC), RMSE, Nash–Sutcliffe efficiency (NSE), and percent bias (PBIAS), and generated quick-look hydrographs.

Script. `scripts/baseline_eval.py`

Deliverables.

- `results/baseline/watauga_cluster_2010_2020_baseline_metrics.csv`
- `results/baseline/watauga_cluster_2010_2020_baseline_hydrograph_<site>.png` (one per site)

1.5 LSTM residual-correction baseline

Process.

- Trained per-site LSTM models for same-time residual correction using past windows only (to prevent information leakage).
- Aggregated per-site metrics to a summary table and generated summary plots.

Scripts.

- `modeling/train_quick_lstm_torch.py`
- `scripts/aggregate_lstm_metrics.py`

Deliverables.

- Per-site metrics and evaluation outputs:
 - `data/clean/modeling/watauga_cluster_2010_2020_lstm_<site>_metrics.json`
 - `data/clean/modeling/watauga_cluster_2010_2020_lstm_<site>_eval.csv`
- Summary table and plots:
 - `results/lstm/watauga_cluster_2010_2020_lstm_summary.csv`
 - `results/lstm/watauga_cluster_2010_2020_lstm_summary_rmse.png`
 - `results/lstm/watauga_cluster_2010_2020_lstm_summary_nse.png`

2 Week 2 (January 28–February 3, 2026): Advanced Models, Ablations, and Tuning

Goal. Evaluate advanced model variants (v2-only), conduct minimal ablations, and perform limited tuning sufficient for a publishable comparison.

2.1 Advanced model training (Hydra v2)

Process.

- Trained Hydra v2 (Transformer–GRU hybrid) per site.
- Assessed performance against CC, RMSE, NSE, and PBIAS.
- For site 03161000, applied post-hoc bias calibration to improve PBIAS without degrading CC/RMSE/NSE.

Scripts.

- `modeling/train_quick_transformer_torch.py`
- `scripts/bias_calibrate_eval.py`

Deliverables.

- Per-site Hydra outputs:
 - `data/clean/modeling/watauga_cluster_2010_2020_hydra_<site>_metrics.json`
 - `data/clean/modeling/watauga_cluster_2010_2020_hydra_<site>_eval.csv`
- Bias-calibrated outputs (site 03161000):
 - `data/clean/modeling/watauga_cluster_2010_2020_hydra_03161000_metrics_biascal.json`
 - `data/clean/modeling/watauga_cluster_2010_2020_hydra_03161000_eval_biascal.csv`

2.2 Ablation study (GRU on/off)

Process.

- Completed Hydra v1 (Transformer-only) runs for all sites.
- Compiled an ablation table comparing Hydra v2 (GRU on) versus Hydra v1 (GRU off).

Script. `scripts/aggregate_hydra_ablation.py`

Deliverable. `results/hydra/watauga_cluster_2010_2020_ablation_gru.csv`

2.3 Targeted tuning

Process.

- Adjusted bias control for 03161000 to ensure CC/RMSE/NSE/PBIAS improve relative to raw NWM.
- Maintained consistent training windows and architecture settings across sites to support fair comparison.

3 Deliverables Summary

3.1 Week 1 deliverables (January 27 target)

- Processed per-site parquets: `data/processed/watauga_cluster_2010_2020_site_<id>.parquet` (4 sites)
- Combined parquet: `data/processed/watauga_cluster_2010_2020_all_sites.parquet`
- Baseline metrics and plots: `results/baseline/watauga_cluster_2010_2020_baseline_metrics.csv`; `results/baseline/watauga_cluster_2010_2020_baseline_hydrograph_<site>.png`
- LSTM summary: `results/lstm/watauga_cluster_2010_2020_lstm_summary.csv` and corresponding RMSE/NSE plots

3.2 Week 2 deliverables (February 3 target)

- Ablation table (GRU on/off): `results/hydra/watauga_cluster_2010_2020_ablation_gru.csv`
- Selected best configuration: Hydra v2 (Transformer-GRU hybrid) with bias calibration for 03161000
- Corrected-versus-raw summary outputs per site:
 - `results/hydra/watauga_cluster_2010_2020_hydra_summary.csv`
 - `results/hydra/watauga_cluster_2010_2020_hydra_summary_rmse.png`
 - `results/hydra/watauga_cluster_2010_2020_hydra_summary_nse.png`
 - `results/hydra/watauga_cluster_2010_2020_hydra_summary_pbias.png`
 - `results/hydra/watauga_cluster_2010_2020_hydra_summary_cc.png`

4 Reproducibility: minimal command set

Build training parquet.

```
.venv/bin/python modeling/build_training_dataset.py \  
  --raw-dir data/raw \  
  --out-dir data/clean/modeling \  
  --start 2010-01-01 --end 2020-12-31 \  
  --sites 03479000 03486000 03161000 03164000 \  
  --nwm-version v2
```

Build processed parquets.

```
.venv/bin/python scripts/build_processed_parquets.py \  
  --data data/clean/modeling/hourly_training_03161000_03164000_03479000_03486000_2010-01-01_2020-12-31 \  
  --output-prefix watauga_cluster_2010_2020
```

Baseline metrics.

```
.venv/bin/python scripts/baseline_eval.py \  
  --data data/clean/modeling/hourly_training_03161000_03164000_03479000_03486000_2010-01-01_2020-12-31 \  
  --output-prefix watauga_cluster_2010_2020
```

LSTM baseline.

```
env PYTHONPATH="$(pwd)" .venv/bin/python modeling/train_quick_lstm_torch.py \  
  --data data/clean/modeling/hourly_training_2010_2020_<site>.parquet \  
  --output-prefix watauga_cluster_2010_2020_lstm_<site>
```

Hydra v2 (hybrid).

```
env PYTHONPATH="$(pwd)" .venv/bin/python modeling/train_quick_transformer_torch.py \  
  --data data/clean/modeling/hourly_training_2010_2020_<site>.parquet \  
  --output-prefix watauga_cluster_2010_2020_hydra_<site>
```

Ablation (Hydra v1 / Transformer-only).

```
env PYTHONPATH="$(pwd)" .venv/bin/python modeling/train_quick_transformer_torch.py \  
  --data data/clean/modeling/hourly_training_2010_2020_<site>.parquet \  
  --output-prefix watauga_cluster_2010_2020_hydra_v1_<site> \  
  --model-arch hydra_v1
```